Mesh Splitter

A mesh splitting and Prefab separation tool for Unity and VRChat. Easily select and isolate any part of a mesh (costumes, accessories, hair, body parts, etc.) into a standalone Prefab and Mesh asset that accurately preserves original transforms, hierarchies, bounds, and necessary components (PhysBones, Constraints, etc.).

Original meshes and avatar assets are never modified.

Quick Start (3 Steps to Split)

Step 1: Open the Tool & Assign Target Mesh

1. Open `dennokoworks > Mesh Splitter` from the top menu bar.
2. Select the object in the Hierarchy and click **"From Selection"** (or drag & drop the Renderer directly into the "Target Mesh" field).

Step 2: Select the Mesh Region

1. Choose a selection unit mode:
   - **UV Island**: Select by continuous UV islands.
   - **Connected Polygons**: Select by 3D connected polygon clusters.
2. Select polygons directly in the **UV Preview** (left-drag to marquee select, right-drag to pan, scroll wheel to zoom) or in the **3D Scene View** (click / drag directly on the mesh).

Step 3: Export as Prefab

1. Enter your desired name in **"Prefab Name"**.
2. Click **"Export Selection as Part Prefab"**.
3. A new Mesh asset and standalone Prefab are generated in the specified folder and automatically placed at the exact original position in your scene.

Screen Layout & Features Reference

1. Top Bar

- **Version Display**: Displays current version and highlights when an update is available.
- **↻ (Recheck Button)**: Manually checks for new releases on GitHub.
- **Language Toggle (`EN` / `JA`)**: Switches the UI display language between English and Japanese (preferences are saved).

2. Target Mesh

Section to configure the source mesh.

- **Target Mesh**: Specify the target SkinnedMeshRenderer or MeshRenderer to split from.
- **From Selection**: Automatically assigns the Renderer currently selected in the Scene / Hierarchy.
- **↻ (Reload)**: Re-analyzes mesh topology and submesh information.

- **Select Submesh**: Limits selection to a specific submesh/material index.
- **UV Channel (`UV0` / `UV1` ...)**: Chooses which UV channel is used to detect UV islands. Only the channels the mesh actually has are listed (`UV0` is fine in most cases).

3. Range Selection

Section to choose which polygons to extract.

- **UV Island**: Selects mesh portions grouped by UV islands.
- **Connected Polygons**: Selects portions connected in 3D mesh space.
- **Select All / Clear / Invert**: Batch operations on the current selection.
- **UV Preview**: Inspect and marquee select on the 2D UV layout. The gray outline marks the 0-1 UV range.
- **Main Texture Display**: When a **submesh is selected** and the **UV channel is `UV0`**, the main texture of that submesh's material is drawn behind the UV layout, so you can see which part of the texture your selection covers.
  - Works with materials that read their main texture (`_MainTex`) from UV0, such as lilToon, Poiyomi and the Unity standard shaders. Tiling and offset are applied, but the texture is drawn only once rather than repeated.
  - The texture is not shown when: the submesh is set to **All**, a channel of `UV1` or later is selected, or a Poiyomi material has its main texture UV set to anything other than `UV0`.
  - UV animation settings such as scrolling and rotation are not reflected.
- **Scene Click Selection (ON/OFF)**: Enables direct clicking and painting on the mesh in the 3D Scene view.
- **Wireframe X-Ray**: Enables see-through wireframe rendering through the object for better visibility.

4. Create Prefab

Section to export the selected mesh as new assets.

- **Prefab Name**: Name of the generated Prefab and Mesh asset.
- **Output Folder**: Destination folder in your project (default: `Assets/MS_splitted_mesh`).
- **Export Selection as Part Prefab**: Exports the selected polygons as a single standalone Prefab and Mesh asset.
- **Batch Export All Submeshes as Individual Prefabs**: Exports all submeshes into separate individual Prefabs in one click.

Key Highlights & Design Safety

- **Zero Alignment Offset**: Original Transform values, bone hierarchies, and localBounds are precisely preserved, so dropping the exported Prefab under its original parent aligns it perfectly.
- **Automatic Component Filtering**: Automatically keeps only the bones, PhysBones, PhysBoneColliders, and Constraints relevant to the extracted mesh. Unrelated avatar components (such as VRC Avatar Descriptor or Animator) will never be mixed in.
- **Non-Destructive**: Never modifies original avatars or source mesh assets.
- **Supports Both Skinned and Static Meshes**: Works seamlessly with SkinnedMeshRenderers (clothes, hair, body) and static MeshRenderers (accessories, props).